

Gradient Flows for Dataset Optimization

Dataset Dynamics via Gradient Flows in Probability Space

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Abstract

Various machine learning tasks, from generative modeling to domain adaptation, revolve around the concept of dataset transformation and manipulation. While various methods exist for transforming unlabeled datasets, principled methods to do so for *labeled* (e.g., classification) datasets are missing. In this work, we propose a novel framework for dataset transformation, which we cast as optimization over data-generating joint probability distributions. We approach this class of problems through Wasserstein gradient flows in probability space, and derive practical and efficient particle-based methods for a flexible but well-behaved class of objective functions. Through various experiments, we show that this framework can be used to impose constraints on classification datasets, adapt them for transfer learning, or to re-purpose fixed or black-box models to classify—with high accuracy—

2019), dataset pooling, or instance mixing (Zhang et al., 2018). On the other hand—and our principal motivation in this work—is the potential of such a framework to tackle less explored or entirely novel problems, such as ‘shaping’ datasets by imposing certain (e.g., geometric or privacy) constraints, or transforming them from one domain to the other to recast transfer learning as transfer of data to the domain of expertise of a model, rather than the other way around. Current data transformation methods are ill suited for these tasks, since they either operate exclusively on features or use labels sparingly and heuristically. In addition, most of them are tailored to specific problems and cannot be easily generalized. Thus, a unified framework for transforming and manipulating datasets is missing.

In response to this, in this work we propose a principled, flexible, and computationally feasible approach to *labeled* dataset transformation. Based on the motivating applications described above, we seek a framework that is model-independent, depending only on intrinsic geometric properties of the data, and applicable to any classification dataset,

Flowing Datasets with Wasserstein over Wasserstein Gradient Flows

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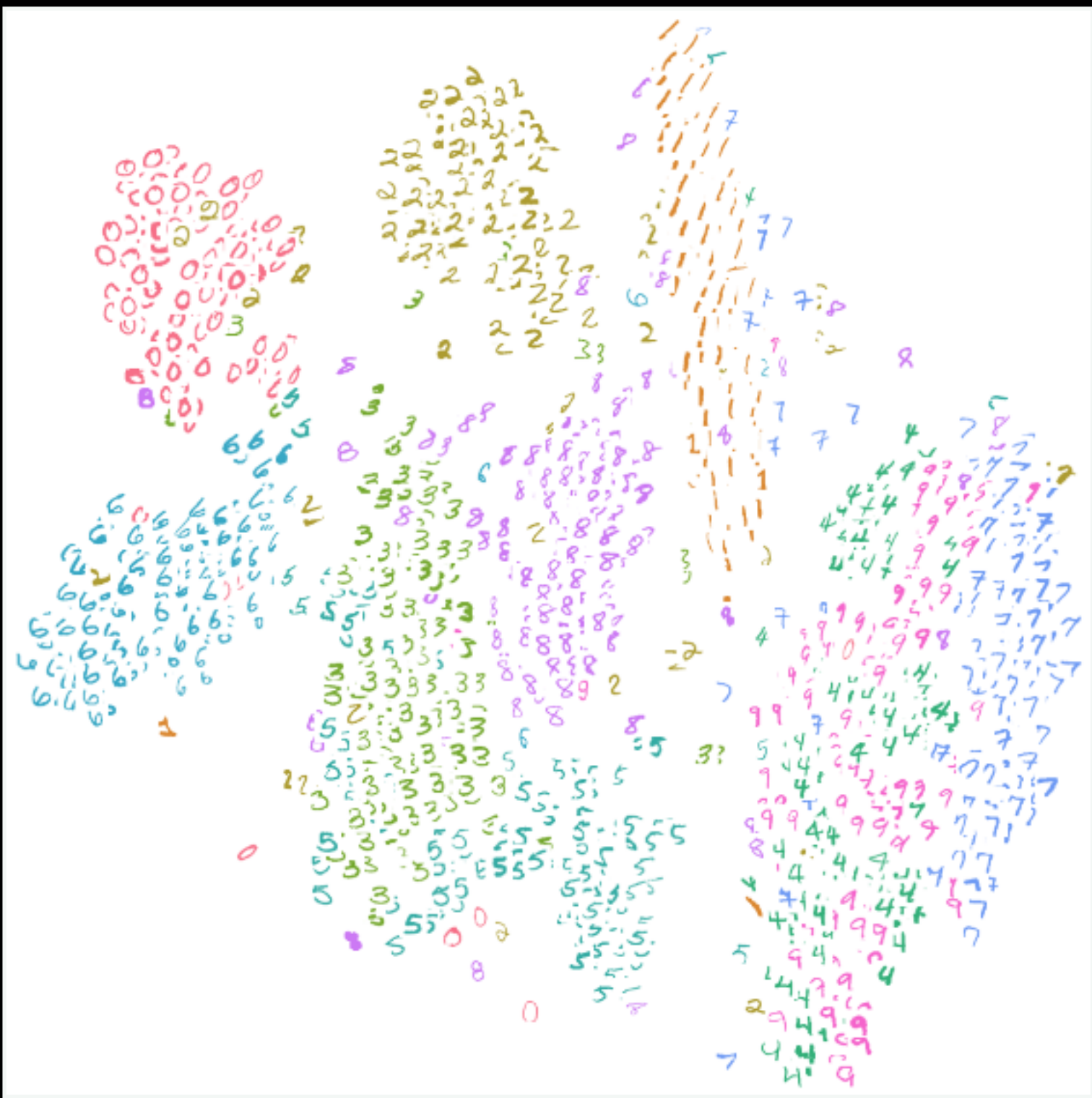
Abstract

Many applications in machine learning involve data represented as probability distributions. The emergence of such data requires radically novel techniques to design tractable gradient flows on probability distributions over this type of (infinite-dimensional) objects. For instance, being able to flow labeled datasets is a core task for applications ranging from domain adaptation to transfer learning or dataset distillation. In this setting, we propose to represent each class by the associated conditional distribution of features, and to model the dataset as a mixture distribution supported on these classes (which are themselves probability distributions), meaning that labeled datasets can be seen as probability distributions over probability distributions. We endow this space with a metric structure from optimal transport, namely the Wasserstein over Wasserstein (WoW) distance,

2023; Haviv et al., 2024b). Remarkably, it has been shown that one can embed any finite dataset with little or no distortion (Andoni et al., 2018; Kratsios et al., 2023) in the Wasserstein space, *i.e.*, the space of probability distributions (e.g., over a Euclidean space) equipped with the Wasserstein-2 distance from Optimal Transport (OT). This has motivated the use of this space to embed many types of data ranging from words (Vilnis & McCallum, 2015) to knowledge graphs (He et al., 2015; Wang et al., 2022), graphs (Bojchevski & Günnemann, 2018; Petric Maretic et al., 2019), or neuroscience data (Bonet et al., 2023). Therefore, it is essential to develop tools to work on the space of probability measures over probability measures, also known as random measures. In particular, they provide a natural way to represent labeled datasets as mixtures (Alvarez-Melis & Fusi, 2020).

A natural distance on this space is the Wasserstein over Wasserstein distance (WoW) (Nguyen, 2016; Catalano & Lavenant, 2024), also known as the Hierarchical OT distance, which lifts the Wasserstein distance between prob-

Motivation: Dataset 'Shaping'



Initial Dataset